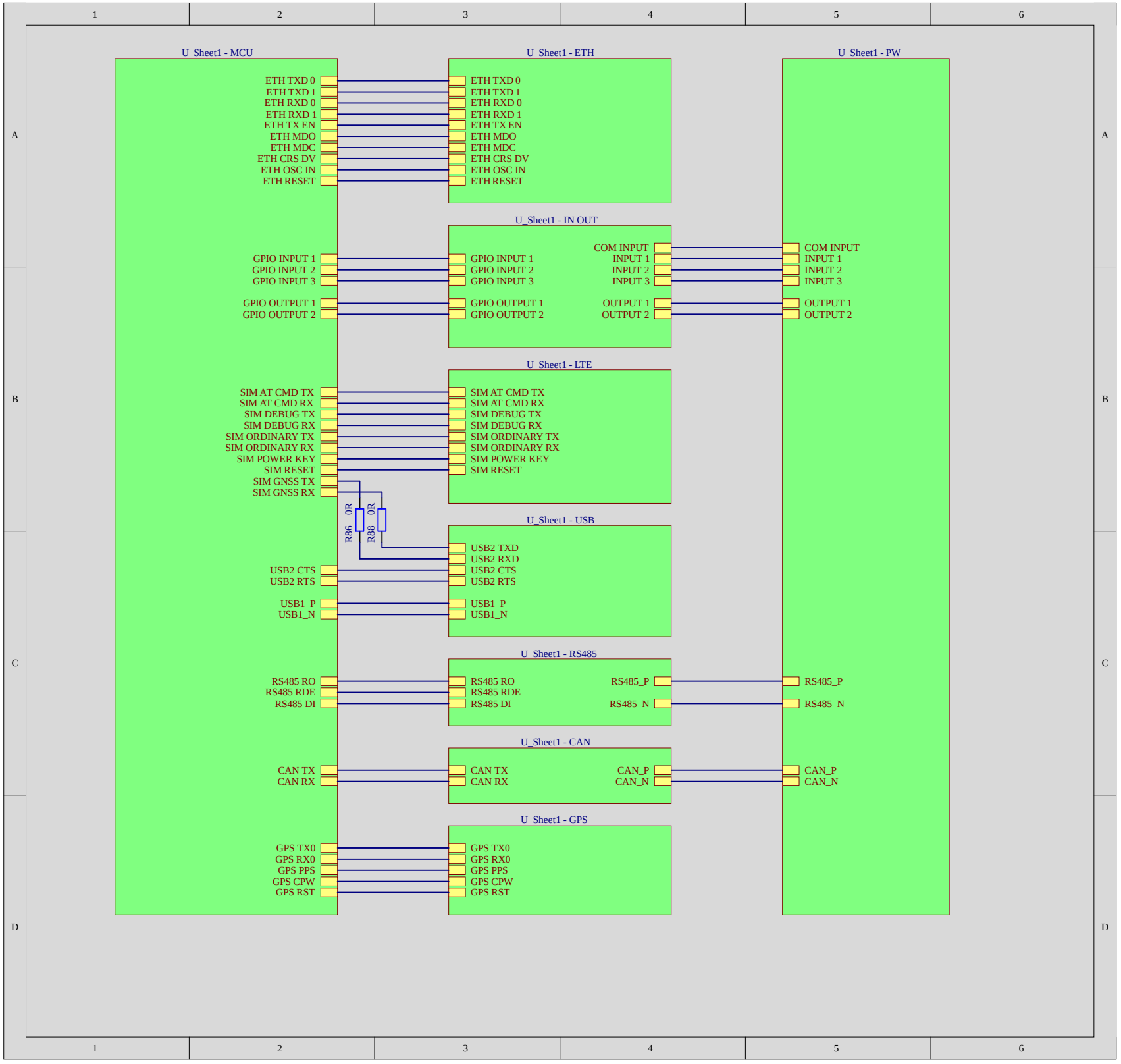
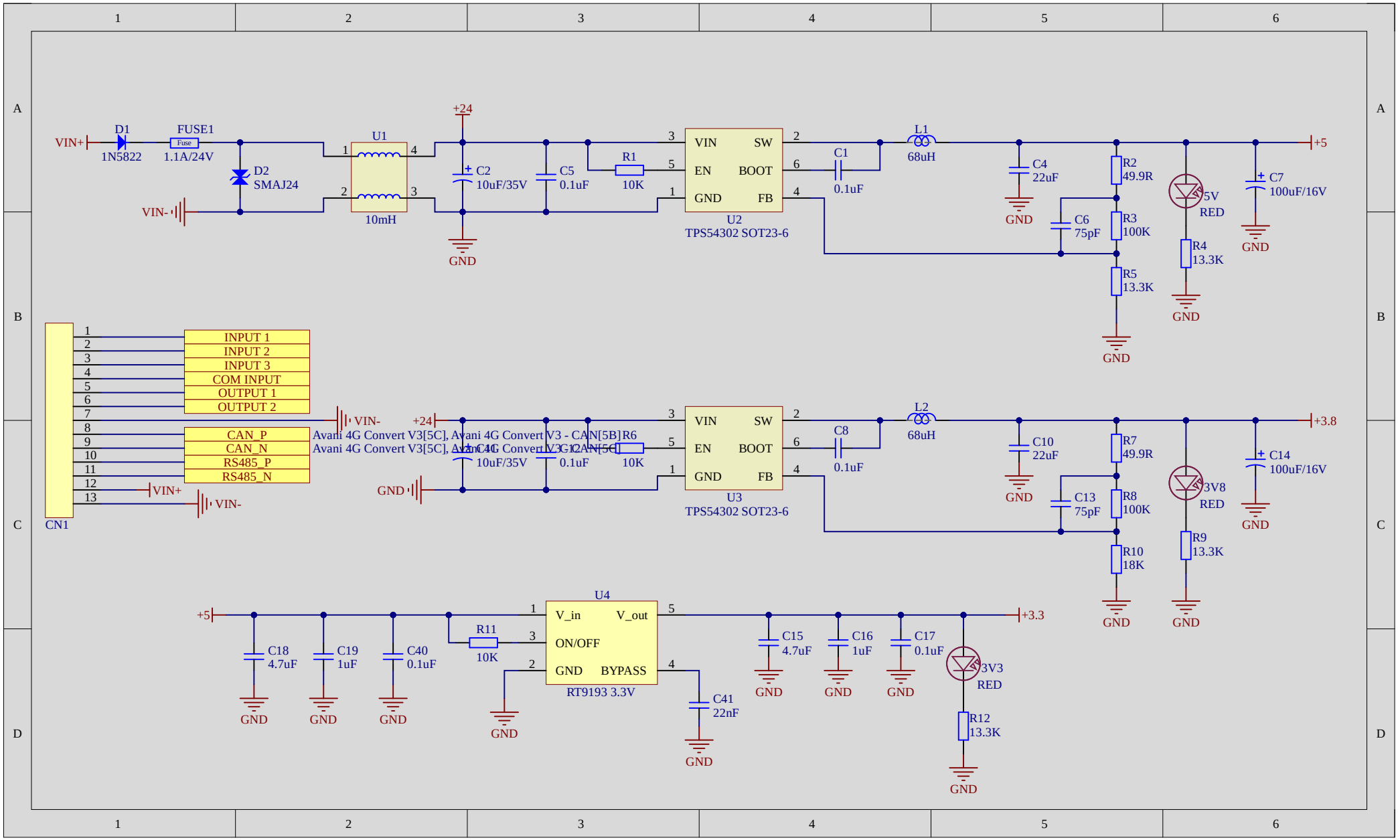
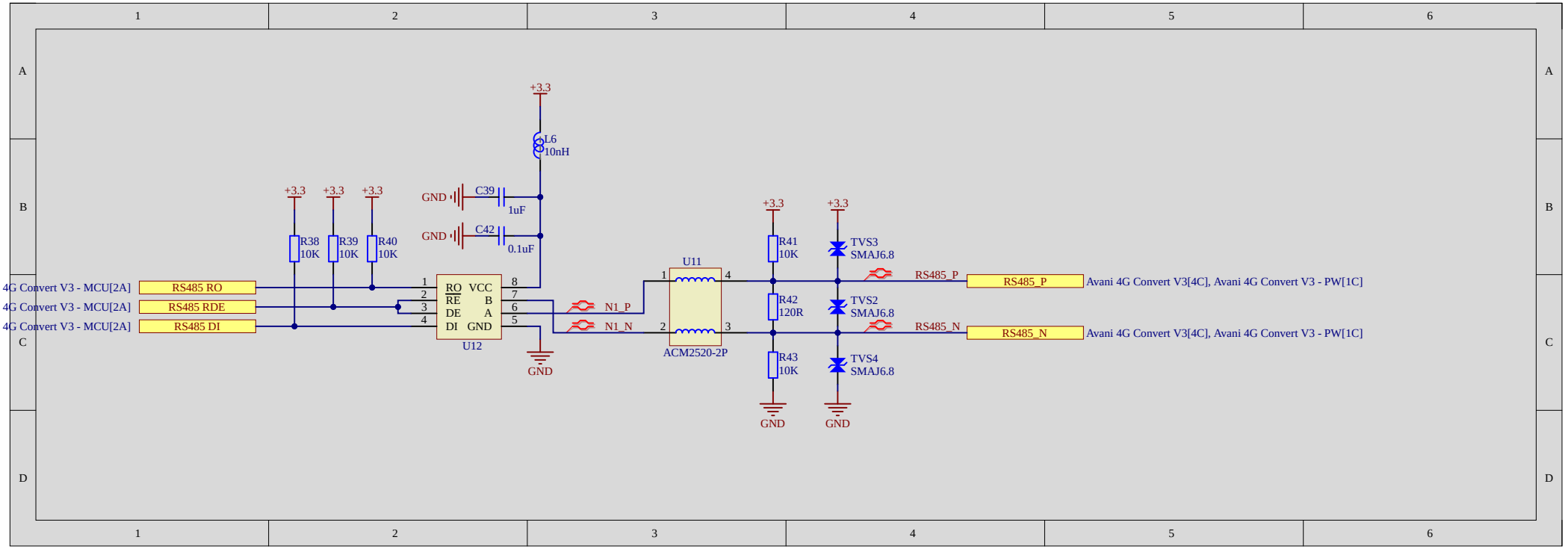
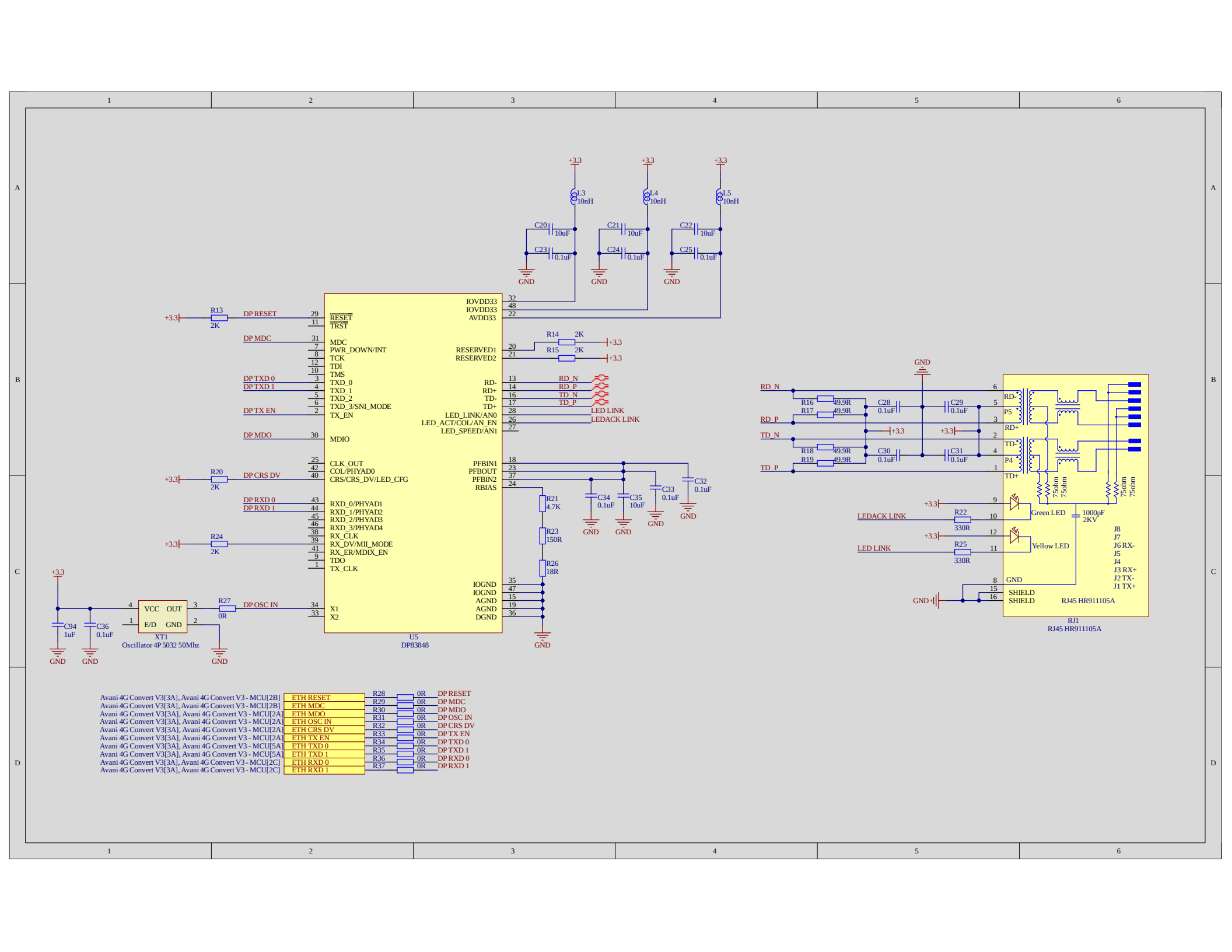
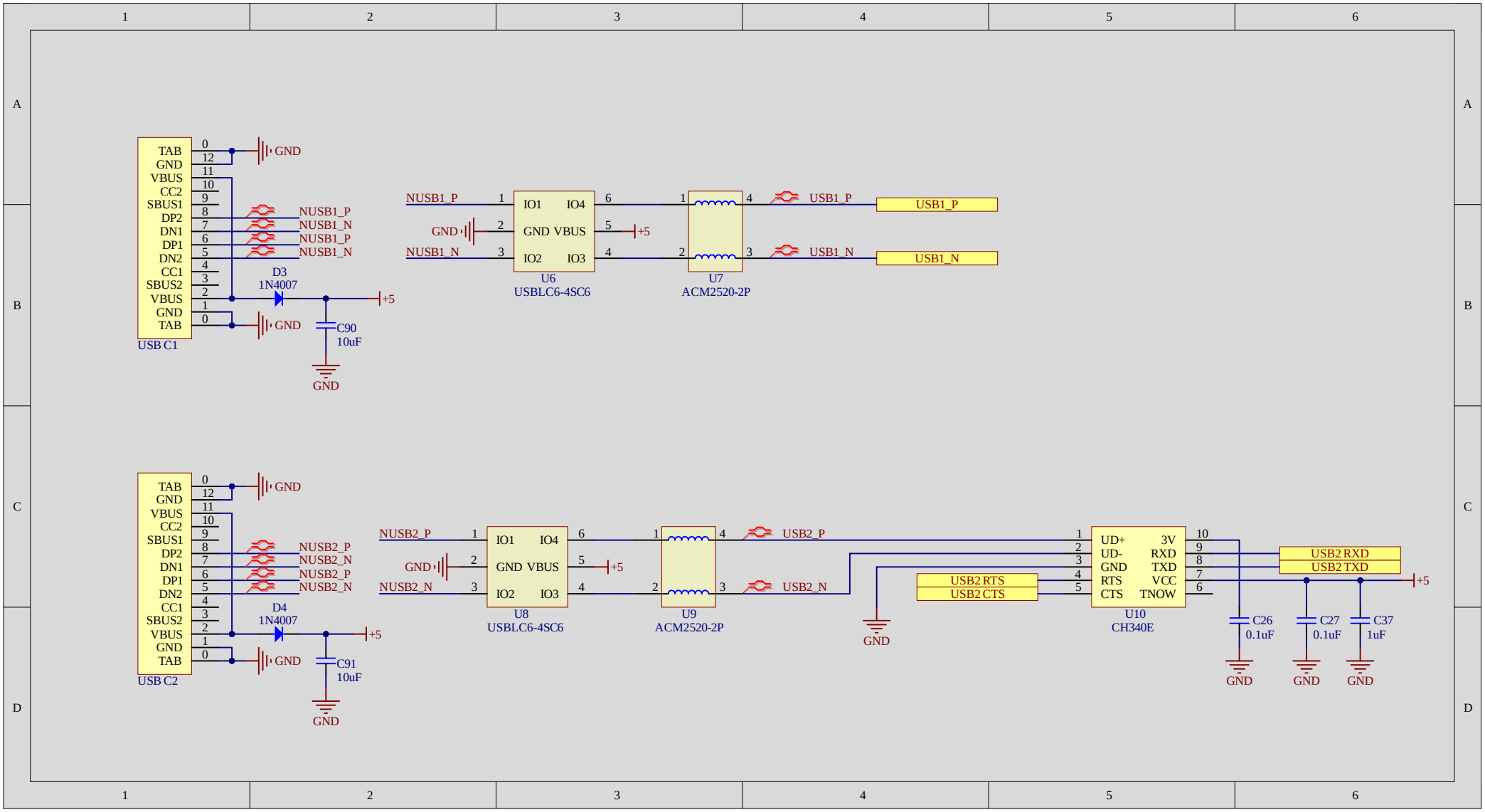
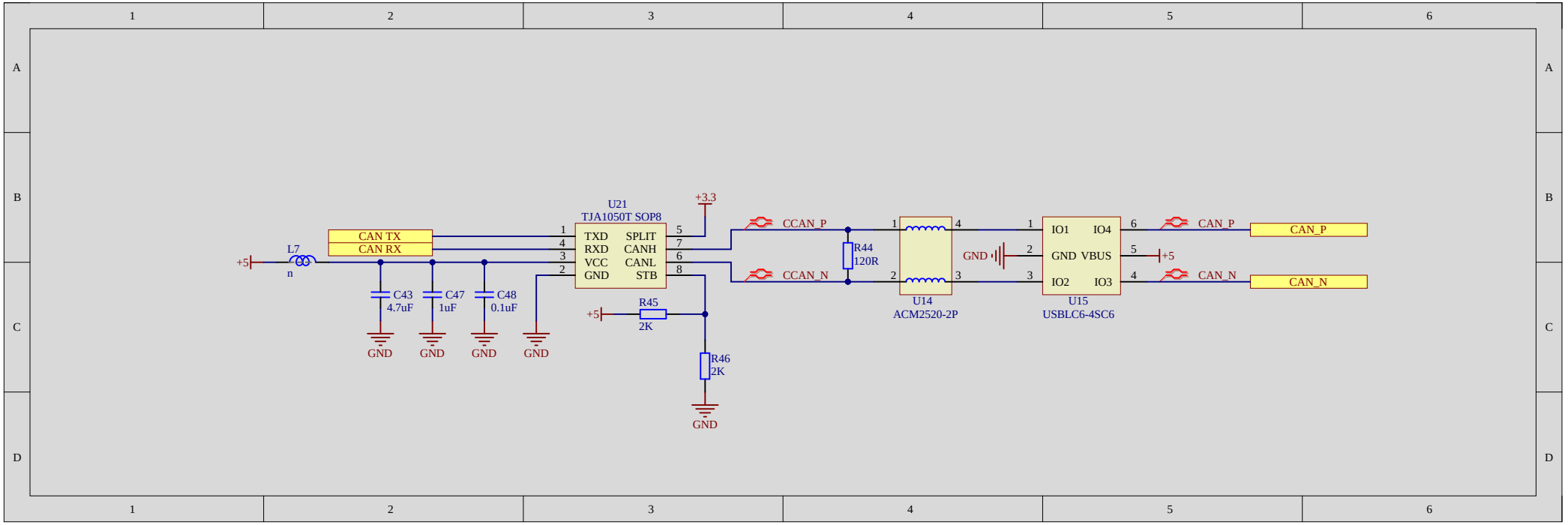

1
2
3
4
5
6
A
B
C
D
U_Sheet1 - MCU
ETH TXD 0
ETH TXD 1
ETH RXD 0
ETH RXD 1
ETH TX EN
ETH MDO
ETH MDC
ETH CRS DV
ETH OSC IN
ETH RESET
GPIO INPUT 1
GPIO INPUT 2
GPIO INPUT 3
GPIO OUTPUT 1
GPIO OUTPUT 2
SIM AT CMD TX
SIM AT CMD RX
SIM DEBUG TX
SIM DEBUG RX
SIM ORDINARY TX
SIM ORDINARY RX
SIM POWER KEY
SIM RESET
SIM GNSS TX
SIM GNSS RX
USB2 CTS
USB2 RTS
USB1_P
USB1_N
RS485 RO
RS485 RDE
RS485 DI
CAN TX
CAN RX
GPS TX0
GPS RX0
GPS PPS
GPS CPW
GPS RST
U_Sheet1 - ETH
ETH TXD 0
ETH TXD 1
ETH RXD 0
ETH RXD 1
ETH TX EN
ETH MDO
ETH MDC
ETH CRS DV
ETH OSC IN
ETH RESET
U_Sheet1 - IN OUT
GPIO INPUT 1
GPIO INPUT 2
GPIO INPUT 3
GPIO OUTPUT 1
GPIO OUTPUT 2
U_Sheet1 - LTE
SIM AT CMD TX
SIM AT CMD RX
SIM DEBUG TX
SIM DEBUG RX
SIM ORDINARY TX
SIM ORDINARY RX
SIM POWER KEY
SIM RESET
R86 0R
R88 0R
U_Sheet1 - USB
USB2 TXD
USB2 RXD
USB2 CTS
USB2 RTS
USB1_P
USB1_N
U_Sheet1 - RS485
RS485 RO
RS485 RDE
RS485 DI
U_Sheet1 - CAN
CAN TX
CAN RX
U_Sheet1 - GPS
GPS TX0
GPS RX0
GPS PPS
GPS CPW
GPS RST
U_Sheet1 - PW
COM INPUT
INPUT 1
INPUT 2
INPUT 3
OUTPUT 1
OUTPUT 2
RS485_P
RS485_N
CAN_P
CAN_N

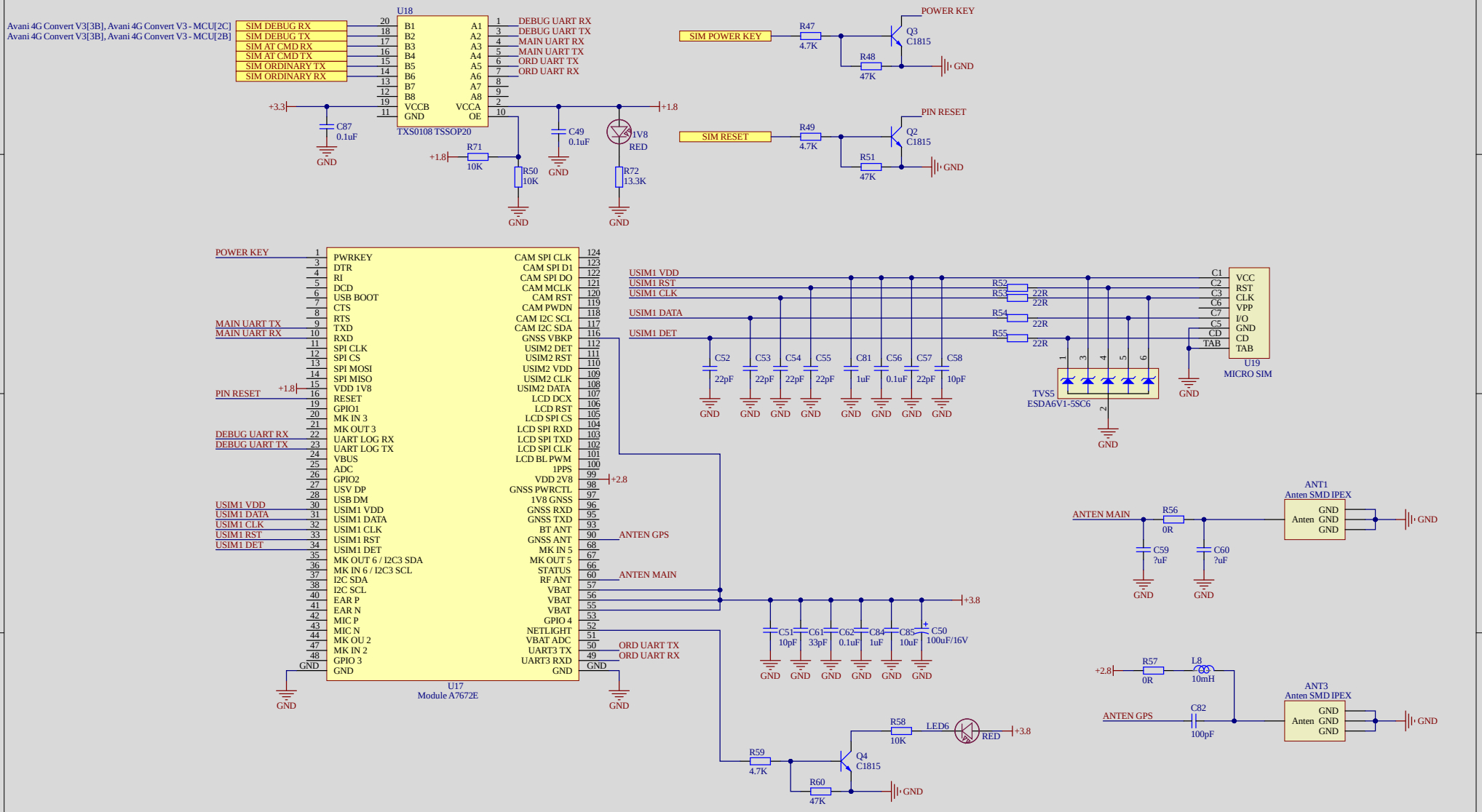

VIN+
D1
1N5822
FUSE1
Fuse
1.1A/24V
D2
SMAJ24
VIN-
U1
10mH
+24
GND
C2
10uF/35V
C5
0.1uF
R1
10K
U2
TPS54302 SOT23-6
VIN
SW
EN
BOOT
GND
FB
C1
0.1uF
L1
68uH
C4
22uF
C6
75pF
R2
49.9R
R3
100K
R5
13.3K
R4
13.3K
5V
RED
C7
100uF/16V
+5
CN1
INPUT 1
INPUT 2
INPUT 3
COM INPUT
OUTPUT 1
OUTPUT 2
CAN_P
CAN_N
RS485_P
RS485_N
Avani 4G Convert V3[5C], Avani 4G Convert V3 - CAN[5B]R6
Avani 4G Convert V3[5C], Avani 4G Convert V3 - CAN[5C]
U3
TPS54302 SOT23-6
C8
0.1uF
L2
68uH
C10
22uF
C13
75pF
R7
49.9R
R8
100K
R10
18K
R9
13.3K
3V8
RED
C14
100uF/16V
+3.8
U4
RT9193 3.3V
V_in
V_out
ON/OFF
BYPASS
R11
10K
C18
4.7uF
C19
1uF
C40
0.1uF
C41
22nF
C15
4.7uF
C16
1uF
C17
0.1uF
3V3
RED
R12
13.3K
+3.3

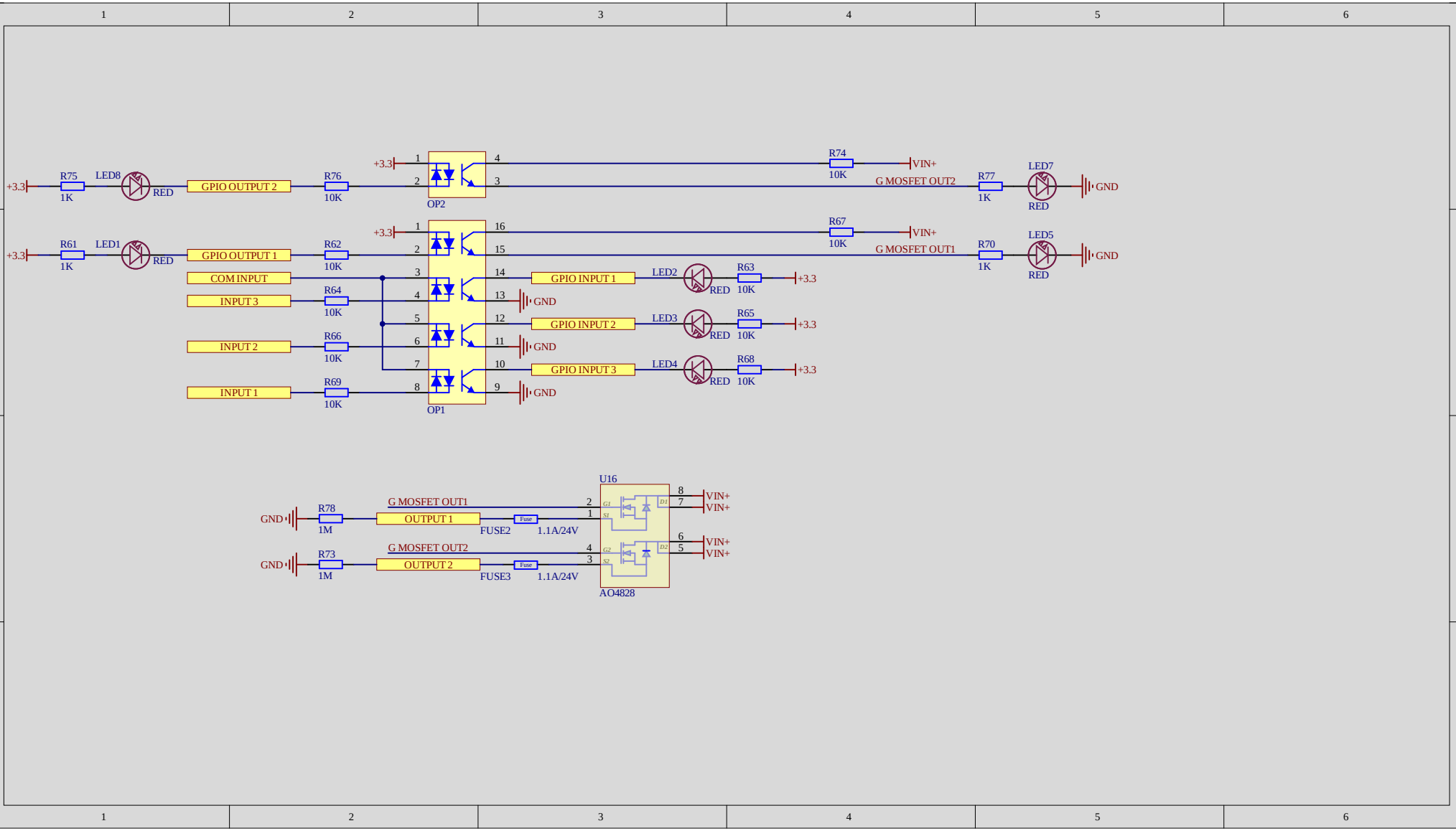

+3.3
L6
10nH
+3.3
+3.3
+3.3
GND
C39
1uF
GND
C42
0.1uF
R38
10K
R39
10K
R40
10K
4G Convert V3 - MCU[2A]
RS485 RO
4G Convert V3 - MCU[2A]
RS485 RDE
4G Convert V3 - MCU[2A]
RS485 DI
1 RO
2 RE
3 DE
4 DI
VCC 8
B 7
A 6
GND 5
U12
GND
N1_P
N1_N
U11
1
2
4
3
ACM2520-2P
+3.3
+3.3
R41
10K
R42
120R
R43
10K
TVS3
SMAJ6.8
TVS2
SMAJ6.8
TVS4
SMAJ6.8
RS485_P
RS485_N
RS485_P
RS485_N
Avani 4G Convert V3[4C], Avani 4G Convert V3 - PW[1C]
Avani 4G Convert V3[4C], Avani 4G Convert V3 - PW[1C]
GND
GND

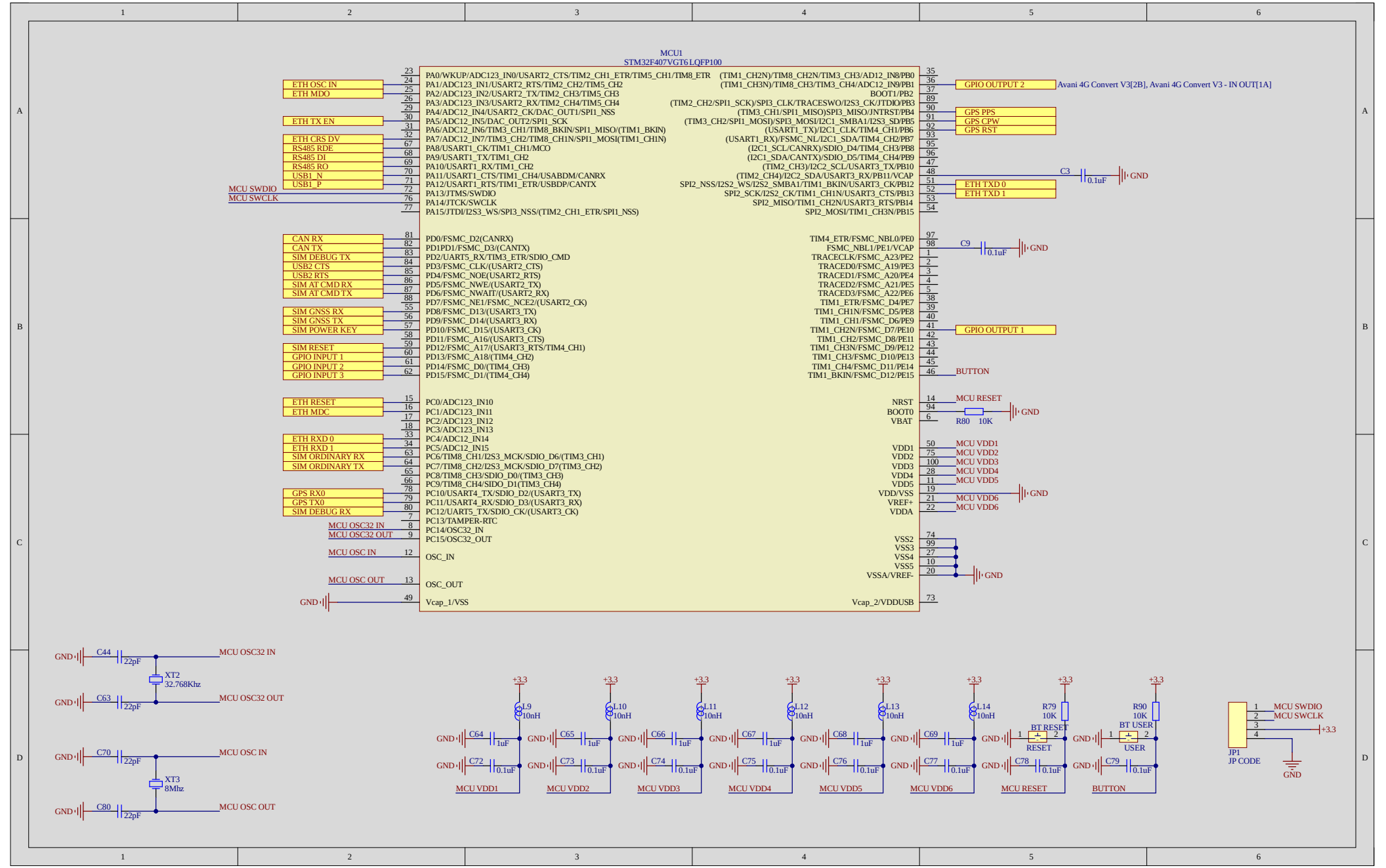

+3.3
L3 10nH
L4 10nH
L5 10nH
C20 10uF
C21 10uF
C22 10uF
C23 0.1uF
C24 0.1uF
C25 0.1uF
GND
R13 2K
DP RESET
R14 2K
R15 2K
DP MDC
DP TXD 0
DP TXD 1
DP TX EN
DP MDO
R20 2K
DP CRS DV
DP RXD 0
DP RXD 1
R24 2K
R27 0R
DP OSC IN
RESET
TRST
MDC
PWR_DOWN/INT
TCK
TDI
TMS
TXD_0
TXD_1
TXD_2
TXD_3/SNI_MODE
TX_EN
MDIO
CLK_OUT
COL/PHYAD0
CRS/CRS_DV/LED_CFG
RXD_0/PHYAD1
RXD_1/PHYAD2
RXD_2/PHYAD3
RXD_3/PHYAD4
RX_CLK
RX_DV/MII_MODE
RX_ER/MDIX_EN
TDO
TX_CLK
X1
X2
IOVDD33
IOVDD33
AVDD33
RESERVED1
RESERVED2
RD-
RD+
TD-
TD+
LED_LINK/AN0
LED_ACT/COL/AN_EN
LED_SPEED/AN1
PFBIN1
PFBOUT
PFBIN2
RBIAS
IOGND
IOGND
AGND
AGND
DGND
RD_N
RD_P
TD_N
TD_P
LED LINK
LEDACK LINK
R21 4.7K
R23 150R
R26 18R
C32 0.1uF
C34 0.1uF
C35 10uF
C33 0.1uF
U5
DP83848
C94 1uF
C36 0.1uF
VCC OUT
E/D GND
XT1
Oscillator 4P 5032 50Mhz
R16 49.9R
R17 49.9R
R18 49.9R
R19 49.9R
C28 0.1uF
C29 0.1uF
C30 0.1uF
C31 0.1uF
RD-
P5
RD+
TD-
P4
TD+
75ohm
Green LED
Yellow LED
1000pF 2KV
R22 330R
R25 330R
LEDACK LINK
LED LINK
GND
SHIELD
SHIELD
J8
J7
J6 RX-
J5
J4
J3 RX+
J2 TX-
J1 TX+
RJ45 HR911105A
RJ1
RJ45 HR911105A
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2B] ETH RESET R28 0R DP RESET
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2B] ETH MDC R29 0R DP MDC
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2A] ETH MDO R30 0R DP MDO
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2A] ETH OSC IN R31 0R DP OSC IN
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2A] ETH CRS DV R32 0R DP CRS DV
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2A] ETH TX EN R33 0R DP TX EN
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[5A] ETH TXD 0 R34 0R DP TXD 0
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[5A] ETH TXD 1 R35 0R DP TXD 1
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2C] ETH RXD 0 R36 0R DP RXD 0
Avani 4G Convert V3[3A], Avani 4G Convert V3 - MCU[2C] ETH RXD 1 R37 0R DP RXD 1

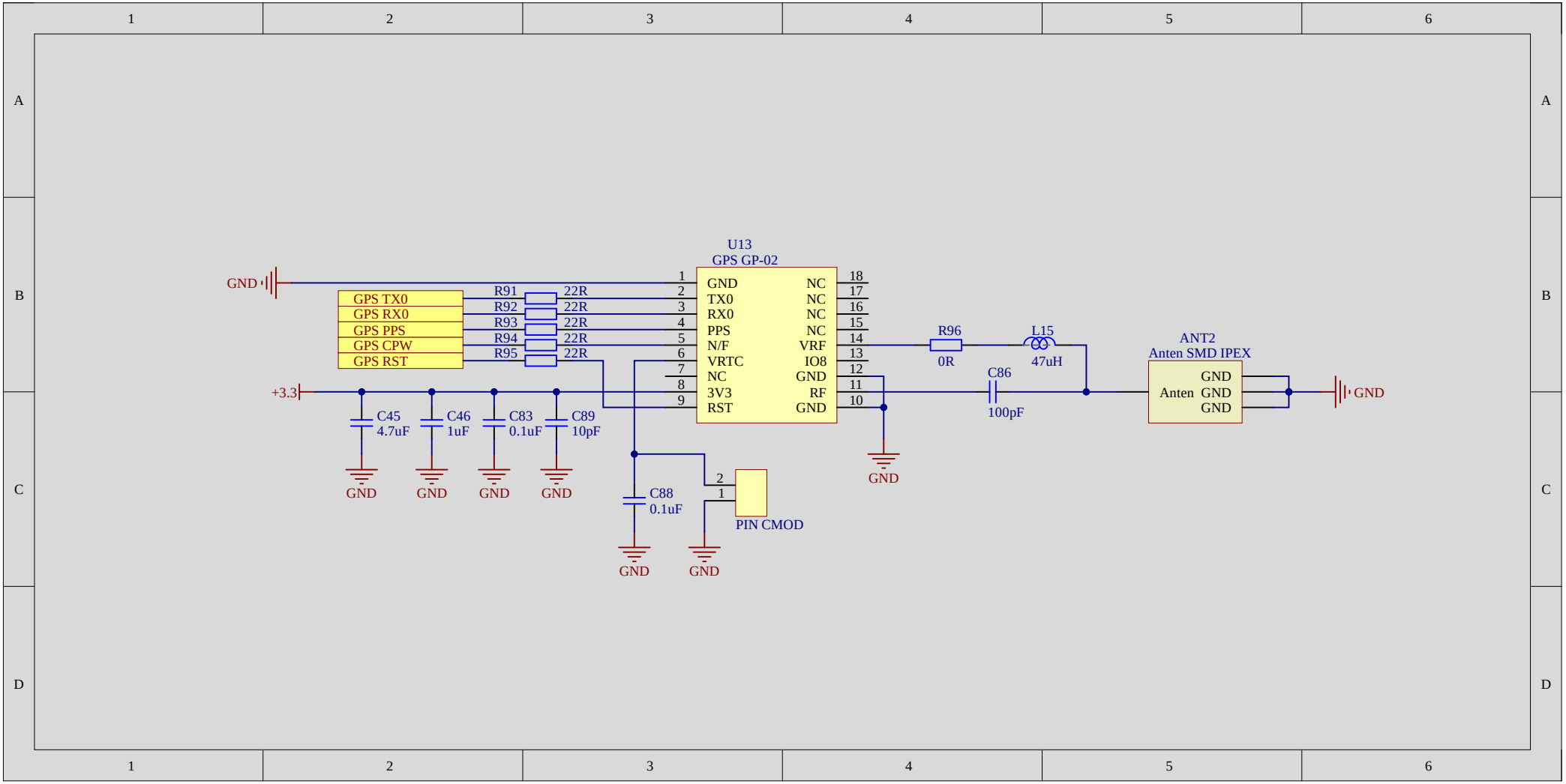

TAB
GND
VBUS
CC2
SBUS1
DP2
DN1
DP1
DN2
CC1
SBUS2
VBUS
GND
TAB
USB C1
NUSB1_P
NUSB1_N
NUSB1_P
NUSB1_N
D3
1N4007
C90
10uF
GND
NUSB1_P
GND
NUSB1_N
IO1
IO4
GND VBUS
IO2
IO3
U6
USBLC6-4SC6
U7
ACM2520-2P
USB1_P
USB1_N
+5
TAB
GND
VBUS
CC2
SBUS1
DP2
DN1
DP1
DN2
CC1
SBUS2
VBUS
GND
TAB
USB C2
NUSB2_P
NUSB2_N
NUSB2_P
NUSB2_N
D4
1N4007
C91
10uF
NUSB2_P
GND
NUSB2_N
U8
USBLC6-4SC6
U9
ACM2520-2P
USB2_P
USB2_N
USB2 RTS
USB2 CTS
UD+
UD-
GND
RTS
CTS
3V
RXD
TXD
VCC
TNOW
U10
CH340E
USB2 RXD
USB2 TXD
C26
0.1uF
C27
0.1uF
C37
1uF


1
2
3
4
5
6
A
B
C
D
CAN TX
CAN RX
L7
n
+5
C43
4.7uF
C47
1uF
C48
0.1uF
GND
U21
TJA1050T SOP8
TXD
RXD
VCC
GND
SPLIT
CANH
CANL
STB
+3.3
R45
2K
R46
2K
CCAN_P
CCAN_N
R44
120R
U14
ACM2520-2P
GND
U15
USBLC6-4SC6
IO1
IO4
GND
VBUS
IO2
IO3
CAN_P
CAN_N


Avani 4G Convert V3[3B], Avani 4G Convert V3 - MCU[2C]
Avani 4G Convert V3[3B], Avani 4G Convert V3 - MCU[2B]
SIM DEBUG RX
SIM DEBUG TX
SIM AT CMD RX
SIM AT CMD TX
SIM ORDINARY TX
SIM ORDINARY RX
U18
TXS0108 TSSOP20
DEBUG UART RX
DEBUG UART TX
MAIN UART RX
MAIN UART TX
ORD UART TX
ORD UART RX
+3.3
+1.8
C87 0.1uF
C49 0.1uF
1V8 RED
R71 10K
R50 10K
R72 13.3K
GND
SIM POWER KEY
R47 4.7K
R48 47K
Q3 C1815
POWER KEY
SIM RESET
R49 4.7K
R51 47K
Q2 C1815
PIN RESET
U17
Module A7672E
PWRKEY
DTR
RI
DCD
USB BOOT
CTS
RTS
TXD
RXD
SPI CLK
SPI CS
SPI MOSI
SPI MISO
VDD 1V8
RESET
GPIO1
MK IN 3
MK OUT 3
UART LOG RX
UART LOG TX
VBUS
ADC
GPIO2
USV DP
USB DM
USIM1 VDD
USIM1 DATA
USIM1 CLK
USIM1 RST
USIM1 DET
MK OUT 6 / I2C3 SDA
MK IN 6 / I2C3 SCL
I2C SDA
I2C SCL
EAR P
EAR N
MIC P
MIC N
MK OU 2
MK IN 2
GPIO 3
CAM SPI CLK
CAM SPI D1
CAM SPI DO
CAM MCLK
CAM RST
CAM PWDN
CAM I2C SCL
CAM I2C SDA
GNSS VBKP
USIM2 DET
USIM2 RST
USIM2 VDD
USIM2 CLK
USIM2 DATA
LCD DCX
LCD RST
LCD SPI CS
LCD SPI RXD
LCD SPI TXD
LCD SPI CLK
LCD BL PWM
1PPS
VDD 2V8
GNSS PWRCTL
1V8 GNSS
GNSS RXD
GNSS TXD
BT ANT
GNSS ANT
MK IN 5
MK OUT 5
STATUS
RF ANT
VBAT
GPIO 4
NETLIGHT
VBAT ADC
UART3 TX
UART3 RXD
ANTEN GPS
ANTEN MAIN
+2.8
+3.8
C51 10pF
C61 33pF
C62 0.1uF
C84 1uF
C85 10uF
C50 100uF/16V
R58 10K
LED6
RED
R59 4.7K
Q4 C1815
R60 47K
USIM1 VDD
USIM1 RST
USIM1 CLK
USIM1 DATA
USIM1 DET
R52 22R
R53 22R
R54 22R
R55 22R
C52 22pF
C53 22pF
C54 22pF
C55 22pF
C81 1uF
C56 0.1uF
C57 22pF
C58 10pF
TVS5
ESDA6V1-5SC6
U19
MICRO SIM
VCC
RST
CLK
VPP
I/O
GND
CD
TAB
ANT1
Anten SMD IPEX
ANTEN MAIN
R56 0R
C59 ?uF
C60 ?uF
R57 0R
L8 10mH
ANT3
Anten SMD IPEX
ANTEN GPS
C82 100pF


1
2
3
4
5
6
A
B
C
D
+3.3
R75
1K
LED8
RED
GPIO OUTPUT 2
R76
10K
OP2
R74
10K
VIN+
G MOSFET OUT2
R77
1K
LED7
RED
GND
R61
1K
LED1
RED
GPIO OUTPUT 1
R62
10K
R67
10K
VIN+
G MOSFET OUT1
R70
1K
LED5
RED
GND
COM INPUT
INPUT 3
R64
10K
INPUT 2
R66
10K
INPUT 1
R69
10K
OP1
GPIO INPUT 1
LED2
R63
10K
GPIO INPUT 2
LED3
R65
10K
GPIO INPUT 3
LED4
R68
10K
U16
R78
1M
G MOSFET OUT1
OUTPUT 1
FUSE2
1.1A/24V
R73
1M
G MOSFET OUT2
OUTPUT 2
FUSE3
1.1A/24V
AO4828
VIN+


MCU1
STM32F407VGT6 LQFP100
PA0/WKUP/ADC123_IN0/USART2_CTS/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR
PA1/ADC123_IN1/USART2_RTS/TIM2_CH2/TIM5_CH2
PA2/ADC123_IN2/USART2_TX/TIM2_CH3/TIM5_CH3
PA3/ADC123_IN3/USART2_RX/TIM2_CH4/TIM5_CH4
PA4/ADC12_IN4/USART2_CK/DAC_OUT1/SPI1_NSS
PA5/ADC12_IN5/DAC_OUT2/SPI1_SCK
PA6/ADC12_IN6/TIM3_CH1/TIM8_BKIN/SPI1_MISO/(TIM1_BKIN)
PA7/ADC12_IN7/TIM3_CH2/TIM8_CH1N/SPI1_MOSI(TIM1_CH1N)
PA8/USART1_CK/TIM1_CH1/MCO
PA9/USART1_TX/TIM1_CH2
PA10/USART1_RX/TIM1_CH2
PA11/USART1_CTS/TIM1_CH4/USABDM/CANRX
PA12/USART1_RTS/TIM1_ETR/USBDP/CANTX
PA13/JTMS/SWDIO
PA14/JTCK/SWCLK
PA15/JTDI/I2S3_WS/SPI3_NSS/(TIM2_CH1_ETR/SPI1_NSS)
(TIM1_CH2N)/TIM8_CH2N/TIM3_CH3/AD12_IN8/PB0
(TIM1_CH3N)/TIM8_CH3/TIM3_CH4/ADC12_IN9/PB1
BOOT1/PB2
(TIM2_CH2/SPI1_SCK)/SPI3_CLK/TRACESWO/I2S3_CK/JTDIO/PB3
(TIM3_CH1/SPI1_MISO)SPI3_MISO/JNTRST/PB4
(TIM3_CH2/SPI1_MOSI)/SPI3_MOSI/I2C1_SMBA1/I2S3_SD/PB5
(USART1_TX)/I2C1_CLK/TIM4_CH1/PB6
(USART1_RX)/FSMC_NL/I2C1_SDA/TIM4_CH2/PB7
(I2C1_SCL/CANRX)/SDIO_D4/TIM4_CH3/PB8
(I2C1_SDA/CANTX)/SDIO_D5/TIM4_CH4/PB9
(TIM2_CH3)/I2C2_SCL/USART3_TX/PB10
(TIM2_CH4)/I2C2_SDA/USART3_RX/PB11/VCAP
SPI2_NSS/I2S2_WS/I2S2_SMBA1/TIM1_BKIN/USART3_CK/PB12
SPI2_SCK/I2S2_CK/TIM1_CH1N/USART3_CTS/PB13
SPI2_MISO/TIM1_CH2N/USART3_RTS/PB14
SPI2_MOSI/TIM1_CH3N/PB15
PD0/FSMC_D2(CANRX)
PD1PD1/FSMC_D3/(CANTX)
PD2/UART5_RX/TIM3_ETR/SDIO_CMD
PD3/FSMC_CLK/(USART2_CTS)
PD4/FSMC_NOE(USART2_RTS)
PD5/FSMC_NWE/(USART2_TX)
PD6/FSMC_NWAIT/(USART2_RX)
PD7/FSMC_NE1/FSMC_NCE2/(USART2_CK)
PD8/FSMC_D13/(USART3_TX)
PD9/FSMC_D14/(USART3_RX)
PD10/FSMC_D15/(USART3_CK)
PD11/FSMC_A16/(USART3_CTS)
PD12/FSMC_A17/(USART3_RTS/TIM4_CH1)
PD13/FSMC_A18/(TIM4_CH2)
PD14/FSMC_D0/(TIM4_CH3)
PD15/FSMC_D1/(TIM4_CH4)
TIM4_ETR/FSMC_NBL0/PE0
FSMC_NBL1/PE1/VCAP
TRACECLK/FSMC_A23/PE2
TRACED0/FSMC_A19/PE3
TRACED1/FSMC_A20/PE4
TRACED2/FSMC_A21/PE5
TRACED3/FSMC_A22/PE6
TIM1_ETR/FSMC_D4/PE7
TIM1_CH1N/FSMC_D5/PE8
TIM1_CH1/FSMC_D6/PE9
TIM1_CH2N/FSMC_D7/PE10
TIM1_CH2/FSMC_D8/PE11
TIM1_CH3N/FSMC_D9/PE12
TIM1_CH3/FSMC_D10/PE13
TIM1_CH4/FSMC_D11/PE14
TIM1_BKIN/FSMC_D12/PE15
PC0/ADC123_IN10
PC1/ADC123_IN11
PC2/ADC123_IN12
PC3/ADC123_IN13
PC4/ADC12_IN14
PC5/ADC12_IN15
PC6/TIM8_CH1/I2S3_MCK/SDIO_D6/(TIM3_CH1)
PC7/TIM8_CH2/I2S3_MCK/SDIO_D7(TIM3_CH2)
PC8/TIM8_CH3/SDIO_D0/(TIM3_CH3)
PC9/TIM8_CH4/SIDO_D1(TIM3_CH4)
PC10/USART4_TX/SDIO_D2/(USART3_TX)
PC11/USART4_RX/SDIO_D3/(USART3_RX)
PC12/UART5_TX/SDIO_CK/(USART3_CK)
PC13/TAMPER-RTC
PC14/OSC32_IN
PC15/OSC32_OUT
OSC_IN
OSC_OUT
Vcap_1/VSS
NRST
BOOT0
VBAT
VDD1
VDD2
VDD3
VDD4
VDD5
VDD/VSS
VREF+
VDDA
VSS2
VSS3
VSS4
VSS5
VSSA/VREF-
Vcap_2/VDDUSB
ETH OSC IN
ETH MDO
ETH TX EN
ETH CRS DV
RS485 RDE
RS485 DI
RS485 RO
USB1_N
USB1_P
MCU SWDIO
MCU SWCLK
CAN RX
CAN TX
SIM DEBUG TX
USB2 CTS
USB2 RTS
SIM AT CMD RX
SIM AT CMD TX
SIM GNSS RX
SIM GNSS TX
SIM POWER KEY
SIM RESET
GPIO INPUT 1
GPIO INPUT 2
GPIO INPUT 3
ETH RESET
ETH MDC
ETH RXD 0
ETH RXD 1
SIM ORDINARY RX
SIM ORDINARY TX
GPS RX0
GPS TX0
SIM DEBUG RX
MCU OSC32 IN
MCU OSC32 OUT
MCU OSC IN
MCU OSC OUT
GND
GPIO OUTPUT 2
Avani 4G Convert V3[2B], Avani 4G Convert V3 - IN OUT[1A]
GPS PPS
GPS CPW
GPS RST
ETH TXD 0
ETH TXD 1
GPIO OUTPUT 1
BUTTON
MCU RESET
MCU VDD1
MCU VDD2
MCU VDD3
MCU VDD4
MCU VDD5
MCU VDD6
C3
0.1uF
C9
0.1uF
R80 10K
C44
22pF
XT2
32.768Khz
C63
22pF
C70
22pF
XT3
8Mhz
C80
22pF
+3.3
L9
10nH
L10
10nH
L11
10nH
L12
10nH
L13
10nH
L14
10nH
C64
1uF
C65
1uF
C66
1uF
C67
1uF
C68
1uF
C69
1uF
C72
0.1uF
C73
0.1uF
C74
0.1uF
C75
0.1uF
C76
0.1uF
C77
0.1uF
R79
10K
BT RESET
RESET
C78
0.1uF
R90
10K
BT USER
USER
C79
0.1uF
JP1
JP CODE


U13
GPS GP-02
GND
GPS TX0
GPS RX0
GPS PPS
GPS CPW
GPS RST
R91 22R
R92 22R
R93 22R
R94 22R
R95 22R
1 GND
2 TX0
3 RX0
4 PPS
5 N/F
6 VRTC
7 NC
8 3V3
9 RST
18 NC
17 NC
16 NC
15 NC
14 VRF
13 IO8
12 GND
11 RF
10 GND
R96
0R
L15
47uH
ANT2
Anten SMD IPEX
GND
Anten GND
GND
GND
C86
100pF
+3.3
C45 4.7uF
C46 1uF
C83 0.1uF
C89 10pF
C88 0.1uF
2
1
PIN CMOD
GND

BT USER
GND
3V3
SWCLK
SWDIO
JP1
ANT1
ANT2
MCU1
BT RESET
Filter Roll UU9.8
CN1
FUSE1
FUSE2
FUSE3
TVS2
TVS3
TVS4
Synapti

Comment	Description	Designator	Footprint	LibRef	Quantity
RED	LED 2P	1V8, 3V3, 3V8, 5V, LED1, LED2, LED3, LED4, LED5, LED6, LED7, LED8	LED 0402 PAD SMALL	LED 2P	12
Anten SMD IPEX	Anten SMD IPEX	ANT1, ANT2, ANT3	Anten SMD IPEX	Anten SMD IPEX	3
RESET	BUTTON SMD 2P 3X4	BT RESET	BUTTON SMD 2P 3X4	BUTTON SMD 2P	1
USER	BUTTON SMD 2P 3X4	BT USER	BUTTON SMD RIGHT 2P 3x6x3.5	BUTTON SMD 2P	1
0.1uF	CAP CERAMIC	C1, C3, C5, C8, C9, C12, C17, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C36, C40, C42, C48, C49, C56, C62, C72, C73, C74, C75, C76, C77, C78, C79, C83, C87, C88	CAP 0402 PAD SMALL	CAP CERAMIC	37
10uF/35V	CAP TANTALUM	C2, C11	CAP TANTALUM SMD 6032	CAP TANTALUM	2
22uF	CAP CERAMIC	C4, C10	CAP 0402 PAD SMALL	CAP CERAMIC	2
75pF	CAP CERAMIC	C6, C13	CAP 0402 PAD SMALL	CAP CERAMIC	2
100uF/16V	CAP TANTALUM	C7, C14	CAP TANTALUM SMD 6032	CAP TANTALUM	2
4.7uF	CAP CERAMIC	C15, C18, C43, C45	CAP 0402 PAD SMALL	CAP CERAMIC	4
1uF	CAP CERAMIC	C16, C19, C37, C39, C46, C47, C64, C65, C66, C67, C68, C69, C81, C84, C94	CAP 0402 PAD SMALL	CAP CERAMIC	15
10uF	CAP CERAMIC	C20, C21, C22, C35, C85, C90, C91	CAP 0402 PAD SMALL	CAP CERAMIC	7
22nF	CAP CERAMIC	C41	CAP 0402 PAD SMALL	CAP CERAMIC	1
22pF	CAP CERAMIC	C44, C52, C53, C54, C55, C57, C63, C70, C80	CAP 0402 PAD SMALL	CAP CERAMIC	9
100uF/16V	CAP TANTALUM	C50	CAP TANTALUM SMD 3528	CAP TANTALUM	1
10pF	CAP CERAMIC	C51, C58, C89	CAP 0402 PAD SMALL	CAP CERAMIC	3
?uF	CAP CERAMIC	C59, C60	CAP 0402 PAD SMALL	CAP CERAMIC	2
33pF	CAP CERAMIC	C61	CAP 0402 PAD SMALL	CAP CERAMIC	1
100pF	CAP CERAMIC	C82, C86	CAP 0402 PAD SMALL	CAP CERAMIC	2
Connector 2EDG 13P 5.08	Connector 2EDG 13P 5.08	CN1	Connector 2EDG 13P 5.08	Connector 2EDG 13P 5.08	1
1N5822	DIODE SCHOTTKY 2P	D1	DIODE SCHOTTKY SMA	DIODE SCHOTTKY 2P	1
SMAJ24	DIODE TVS SMD 2P	D2	DIODE TVS SMA	DIODE TVS SMD 2P	1
1N4007	DIODE SCHOTTKY 2P	D3, D4	DIODE SCHOTTKY SOD123	DIODE SCHOTTKY 2P	2
1.1A/24V	FUSE	FUSE1, FUSE2, FUSE3	FUSE 1812 PAD MEDIUM	FUSE	3
JP CODE	JUMP DIP 2.54 4P-X1	JP1	JUMP DIP 2.54 4P-X1	JUMP DIP 2.54 4P-X1	1
68uH	INDUCTOR SMD	L1, L2	INDUCTOR SMD 3.5X3.2	INDUCTOR SMD	2
10nH	INDUCTOR SMD	L3, L4, L5, L6, L9, L10, L11, L12, L13, L14	INDUCTOR 0402 PAD SMALL	INDUCTOR SMD	10
n	INDUCTOR SMD	L7	INDUCTOR 0402 PAD SMALL	INDUCTOR SMD	1
10mH	INDUCTOR SMD	L8	INDUCTOR 0402 PAD SMALL	INDUCTOR SMD	1
47uH	INDUCTOR SMD	L15	INDUCTOR 0402 PAD SMALL	INDUCTOR SMD	1
STM32F407VGT6 LQFP100	STM32F407VGT6 LQFP100	MCU1	STM32F407VGT6 LQFP100	STM32F407VGT6 LQFP100	1
TLP290-4 SOP16	TLP290-4 SOP16	OP1	TLP290-4 SOP16	TLP290-4 SOP16	1
TLP290 SOP4	TLP290 SOP4	OP2	TLP290 SOP4	TLP290 SOP4	1
JUMP DIP 2.0 2P-X1	JUMP DIP 2.0 2P-X1	PIN CMOD	JUMP DIP 2.0 2P-X1	JUMP DIP 2.0 2P-X1	1
C1815	Transistor NPN	Q2, Q3, Q4	C1815 SOT23	Transistor NPN	3
10K	RES SMD	R1, R6, R11, R38, R39, R40, R41, R43, R50, R58, R62, R63, R64, R65, R66, R67, R68, R69, R71, R74, R76, R79, R80, R90	RES 0402 PAD SMALL	RES SMD	24
49.9R	RES SMD	R2, R7, R16, R17, R18, R19	RES 0402 PAD SMALL	RES SMD	6
100K	RES SMD	R3, R8	RES 0402 PAD SMALL	RES SMD	2
13.3K	RES SMD	R4, R5, R9, R12, R72	RES 0402 PAD SMALL	RES SMD	5
18K	RES SMD	R10	RES 0402 PAD SMALL	RES SMD	1
2K	RES SMD	R13, R14, R15, R20, R24, R45, R46	RES 0402 PAD SMALL	RES SMD	7
4.7K	RES SMD	R21, R47, R49, R59	RES 0402 PAD SMALL	RES SMD	4
330R	RES SMD	R22, R25	RES 0402 PAD SMALL	RES SMD	2
150R	RES SMD	R23	RES 0402 PAD SMALL	RES SMD	1
18R	RES SMD	R26	RES 0402 PAD SMALL	RES SMD	1
0R	RES SMD	R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R56, R57, R86, R88, R96	RES 0402 PAD SMALL	RES SMD	16
120R	RES SMD	R42, R44	RES 0402 PAD SMALL	RES SMD	2
47K	RES SMD	R48, R51, R60	RES 0402 PAD SMALL	RES SMD	3
22R	RES SMD	R52, R53, R54, R55, R91, R92, R93, R94, R95	RES 0402 PAD SMALL	RES SMD	9
1K	RES SMD	R61, R70, R75, R77	RES 0402 PAD SMALL	RES SMD	4
1M	RES SMD	R73, R78	RES 0402 PAD SMALL	RES SMD	2
RJ45 HR911105A	RJ45 HR911105A	RJ1	RJ45 HR911105A	RJ45 HR911105A	1
SMAJ6.8	DIODE TVS SMD 2P	TVS2, TVS3, TVS4	DIODE TVS SOD123	DIODE TVS SMD 2P	3
ESDA6V1-5SC6	ESDA6V1-5SC6 SOT23-6	TVS5	ESDA6V1-5SC6 SOT23-6	ESDA6V1-5SC6 SOT23-6	1
10mH	Filter Roll UU9.8	U1	Filter Roll UU9.8	Filter Roll UU9.8	1
TPS54302 SOT23-6	TPS54302 SOT23-6	U2, U3	TPS54302 SOT23-6	TPS54302 SOT23-6	2
RT9193 3.3V	RT9193 SOT23-5	U4	RT9193 SOT23-5	RT9193 SOT23-5	1
DP83848	DP83848	U5	DP83848 LQFP48	DP83848	1
USBLC6-4SC6	USBLC6-4SC6 SOT23-6	U6, U8, U15	USBLC6-4SC6 SOT23-6	USBLC6-4SC6 SOT23-6	3
ACM2520-2P	Filter Roll ACM2520-2P	U7, U9, U11, U14	Filter Roll ACM2520-2P	Filter Roll ACM2520-2P	4
CH340E	CH340E MSOP10	U10	CH340E MSOP10	CH340E MSOP10	1
MAX485 SOP8	MAX485 SOP8	U12	MAX485 SOP8	MAX485 SOP8	1
GPS GP-02	Module GPS GP-02	U13	Module GPS GP-02	Module GPS GP-02	1
AO4828	AO4828 SOP8	U16	AO4828 SOP8	AO4828 SOP8	1
Module A7672E	Module A7672E	U17	Module A7672E	Module A7672E	1
TXS0108 TSSOP20	TXS0108 TSSOP20	U18	TXS0108 TSSOP20	TXS0108 TSSOP20	1
MICRO SIM	SOCKET MICRO SIM 7P	U19	SOCKET MICRO SIM 7P	SOCKET MICRO SIM 7P	1
TJA1050T SOP8	TJA1050T SOP8	U21	TJA1050T SOP8	TJA1050T SOP8	1
SOCKET USB TYPE-C SMD 16PIN RIGHT	SOCKET USB TYPE-C SMD 16PIN RIGHT	USB C1, USB C2	SOCKET USB TYPE-C SMD 16PIN RIGHT	SOCKET USB TYPE-C SMD 16PIN RIGHT	2
Oscillator 4P 5032 50Mhz	CRYSTAL 4P 5032	XT1	CRYSTAL 4P 5032	CRYSTAL 4P 5032	1
32.768Khz	CRYATAL 2P	XT2	CRYSTAL 2P FC-135	CRYATAL 2P	1
8Mhz	CRYATAL 2P	XT3	CRYSTAL 2P 5032	CRYATAL 2P	1